Engineering Physics for CSE Stream

Module 2: Quantum Mechanics

CONTENTS

Quantum Mechanics: de Broglie Hypothesis and Matter Waves, de Broglie wavelength and derivation of expression by analogy, Phase Velocity and Group Velocity, Heisenberg's Uncertainty Principle and its application (Non-existence of electron inside the nucleus - Non Relativistic), Principle of Complementarity, Wave Function, Time independent Schrödinger wave equation (Derivation), Physical Significance of a wave function and Born Interpretation, Expectation value, Eigen functions and Eigen Values, Particle inside one dimensional infinite potential well, Quantization of Energy States, Waveforms and Probabilities. Numerical Problems.

Important Constant values:

Planck's Constant: $h = 6.625 \times 10^{-34}$ J-s

Charge on electron (q or e) = 1.6×10^{-19}

Mass of electron $m = 9.1 \times 10^{-31}$ kg

Speed of Light in vacuum $c = 3 \times 10^{8}$ m/s

For numerical problems based on;

1. **De-Broglie wavelength.**
2. **Uncertainty Principle.**
3. **Probability of finding the particle in 1-dimenstion.**
4. **Energy Eigen value for particle in an infinite potential well (Box)**
5. **Expectation Value.**

Dual nature of light:

The evidence supporting the particle nature of light is provided by the concepts of the photoelectric effect and the Compton effect. In contrast, physical optics used the wave nature of light to explain phenomena like interference, diffraction, and superposition. This is the duality of light waves and particles.

Dual nature of matter:

On the basis of dual nature of light, in 1924, Louis de Broglie suggested that all the microscopic material particles such as electrons, protons, atoms, molecules etc. also exhibit

*"**If the particle of matter has a momentum then waves associated with it**"*, this is known as **de Broglie hypothesis.** Therefore, some sort of waves propagates with moving material particles called Matter waves or De-Broglie waves and ***the wavelength associated with matter waves is called de Broglie wavelength.***

Its relation is given by $\lambda = \frac{h}{p}$

Where h= 6.6X10^{-34} JS is called Planck's constant and p =mv is the momentum, "m' is the mass of the moving particle and 'v' is its velocity.

Matter is made up of discrete constituent particles like atoms, molecules, protons, neutrons and electrons, hence matter has particle nature. Wave nature of matter is explained by Davisson and Germer experiment. Hence matter also exhibit wave particle duality.

de Broglie wavelength derivation by analogy and its different forms:

A particle of mass 'm' moving with velocity 'c' possess energy given by Einstein's Equation

$$E = mc^2 \rightarrow (1)$$

According to Planck's quantum theory the energy of quantum of frequency 'υ' is

$$E = h\upsilon \rightarrow (2)$$

From (1) & (2) $mc^2 = h\upsilon = hc/\lambda$ (since $\upsilon = c/\lambda$)

$$\lambda = hc/mc^2 = h/mc$$

Since v ≈ c,

$$\therefore \boldsymbol{\lambda = \frac{h}{mv} = \frac{h}{p}} \quad \rightarrow (\mathbf{1})$$

Where p is the momentum of the particle.

de Broglie wavelength of a free particle in terms of its kinetic energy

Consider a particle, since the particle is free, the total energy is same as kinetic energy:

$$E_k = \frac{1}{2}\, mv^2$$

Multiply ✕ and divide ➗ by m on the right-side term.

$$E_k = \frac{1}{2m}\, m^2 v^2$$

$$E_k = \frac{p^2}{2m} \quad since\ p = mv$$

$$p^2 = \ 2mE_k$$

$$p = \sqrt{2mE_k} \ \longrightarrow (\mathbf{2})$$

Substitute the value of p in de Broglie's formula

$$\Longrightarrow \quad \boldsymbol{\lambda = \frac{h}{\sqrt{2mE_k}}} \ \longrightarrow (\mathbf{3})$$

de Broglie Wavelength of an Accelerated Electron:

Now, consider a charged particle of charge *q*, mass *m* accelerated through a potential difference (voltage) *V*. The work done on this particle due to the potential difference is given by *qV*. This work done is converted into the particle's kinetic energy, i.e., $E_k = qV$. Substitute this E_k in an equation (3).

$$\lambda = \frac{h}{\sqrt{2mqV}} \longrightarrow (4)$$

Here, q is the magnitude of the charge.

If an electron accelerated with potential difference 'V' then the work done on this particle due to the potential difference is 'eV', Then eqn. (4) becomes,

$$\lambda = \frac{h}{\sqrt{2meV}}$$

After rearranging,

$$\lambda = \frac{h}{\sqrt{2me}} \times \left(\frac{1}{\sqrt{V}}\right) \longrightarrow (4)$$

We know that, Planck's Constant: $h = 6.625 \times 10^{-34}$ *J-s*, Charge on electron, $e = 1.6 \times 10^{-19}$, Mass of electron, $m = 9.1 \times 10^{-31}$ *kg.* Then eqn. (5) becomes,

$$\lambda = \frac{6.625 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 1.6 \times 10^{-19}}} \times \left(\frac{1}{\sqrt{V}}\right)$$

$$\boldsymbol{\lambda = \frac{12.27}{\sqrt{V}}} \text{ Å} \longrightarrow (\mathbf{5})$$

Equations (1), (3) and (5) are different forms of de Broglie's wavelength of particle.

Matter waves and its properties (Characteristics):

1. Phase velocity $\boldsymbol{v_p}$:

The velocity with which a wave travels is called phase velocity and is also called wave velocity. It is given by

$$v_p = \nu\lambda = \frac{\omega}{k} = \frac{c^2}{v} \longrightarrow (6)$$

If a point is marked on the wave representing the phase of the particle then the velocity with which the phase propagates from one point to another is called phase velocity.

Here is the 'c' velocity of light in vacuum and 'v, is the velocity of the matter wave in the last term of the equation. Since, phase velocity is greater than the velocity of light. Hence there is no physical significance for phase velocity of matter waves.

2. Wave packet and group velocity v_g:

Since the velocity of matter waves must be equal to that of the particle velocity and since no physical meaning can be associated with phase velocity, the concept of group velocity is introduced.

Wave packet *is the resultant wave of matter waves which is formed due to the superposition of two or more waves with slightly differ in their frequencies or velocities.*

The velocity with which the wave packet propagates is called group velocity which is same as particle velocity. It is denoted by v_g and is as given by

$$v_g = \frac{d\omega}{dk} \longrightarrow (7)$$

Characteristics (Properties) of matter waves:

1. Waves associated with moving particles are called matter waves.

2. The wavelength 'λ' of a de Broglie wave associated with particle of mass 'm' moving with velocity 'v' is λ = h/(mv)

2. Matter waves are not electromagnetic waves because the de Broglie wavelength is independent of charge of the moving particle.

3. The amplitude of the matter wave depends on the probability of finding the particle in that position.

4. The speed of matter waves depends on the mass and velocity of the particle associated with the wave.

5. The phase velocity has no physical meaning for matter waves

6. Group velocity is associated with matter waves.

Heisenberg's Uncertainty Principle:

According to classical mechanics a particle occupies a definite place in space and possesses a definite momentum. If the position and momentum of a particle is known at any instant of time, it is possible to calculate its position and momentum at any later instant of time. The path of the particle could be traced. This concept breaks down in quantum mechanics leading to Heisenberg's Uncertainty Principle.

Heisenberg's Uncertainty Principle states that ***"It is impossible to measure simultaneously both the position and momentum of a particle accurately".***

According to the principle, in any simultaneous determination of position and momentum of the particle, the product of uncertainties is equal to or greater than $\frac{h}{4\pi}$.

$$\boldsymbol{\Delta x \,.\, \Delta p \geq \frac{h}{4\pi}} \longrightarrow (\mathbf{8})$$

Where $\boldsymbol{\Delta x}$ is the error (or uncertainty) in the measurement of position and $\boldsymbol{\Delta p}$ is the error (or uncertainty) in the measurement of momentum. The product of the errors is of the order of Planck's constant.

Principle of Complementarity

Statement: Principle of complementarity as stated by Bohr "In a situation where the wave aspect of the system is revealed, its particle aspect is concealed (hidden) and in a situation where the particle aspect is revealed its wave aspect is concealed (hidden). Revealing both simultaneously is impossible; the wave and aspects are complementary."

(**Note:** Meaning of complementary: things are different from each other but make a good combination.)

The complementarity principle holds that objects have certain pairs of complementary properties which cannot be observed or measured *accurately* simultaneously. One of these pairs is position and momentum. Other pairs include

1. Energy (E) and time (t) with $\boldsymbol{\Delta E} \,.\, \boldsymbol{\Delta t} \geq \frac{\boldsymbol{h}}{\boldsymbol{4\pi}}$

2. Angular displacement (θ) and angular momentum (L) with $\boldsymbol{\Delta \theta} \,.\, \boldsymbol{\Delta L} \geq \frac{\boldsymbol{h}}{\boldsymbol{4\pi}}$

Significance of Heisenberg's Uncertainty Principle

Heisenberg's Uncertainty Principle asserts that it is impossible to measure simultaneously both the position and momentum of a particle accurately. If we make an effort to measure very accurately the position of a particle, it leads to large uncertainty in the measurement of momentum and vice versa. Therefore, one should think only of the probability of finding the particle at a certain position or of the probable value for the momentum of the particle.

Application of Heisenberg's Uncertainty Principle

Non-existence of electrons in the atomic nucleus:

In the β-decay process of radioactive radiation, it was observed that the emitted radiation from the nucleus was composed of electrons (β rays), and the energy of these electrons was found to be in the range of 3 to 4 *MeV*. If the electron can exist inside the atomic nucleus then uncertainty in its position must not exceed the diameter of the nucleus.

Therefore, the diameter of the nucleus is of the order of $\Delta x = 10^{-14}\, m$

Applying Heisenberg's uncertainty principle for an electron expected to be inside the nucleus.

We have $$\Delta x \,.\, \Delta p \geq \frac{h}{4\pi}$$

$$\Delta p \geq \frac{h}{4\pi\, \Delta x}$$

$$\Delta p = \frac{6.625 \times 10^{-34}}{4 \times 3.142 \times 10^{-14}}$$

$$\Delta p = 0.527 \times 10^{-20} = 5.27 \times 10^{-21}$$

Therefore, the electron should possess momentum ≃ Δp = 5.27×10^{-21}

Non-relativistic equation of energy of the electron is given by

$$E = \frac{1}{2}mv^2 = \frac{p^2}{2m} = \frac{(5.27 \times 10^{-21})^2}{2 \times 9.1 \times 10^{-31}} \geq 1.525 \times 10^{-11}\, J$$

here m is the rest mass of the electron = $9.1 \times 10^{-31} kg$

$$\boldsymbol{E \geq \frac{1.525 \times 10^{-11}}{1.6 \times 10^{-19}} \geq 95\ MeV}$$

Conclusion: According to experiments, the energy associated with the beta ray (electron) emission is around 3-4 MeV which is much lesser than the energy of the electron expected to be inside the nucleus 93 MeV. Hence electrons cannot exist inside the nucleus.

Wave Function:

A physical situation in quantum mechanics is represented by a function called wave function. It is denoted by 'ψ'. It accounts for the wave like properties of particles. Wave function is obtained by solving Schrodinger equation. Mathematically it is given by $\psi = Ae^{i(kx-\omega t)}$. It can be express in complex form. A wave function that depends on space (x, y and z) and time (t) and is denoted by ψ (r, t).

Physical significance of wave function:

The wave function itself has no physical significance, the physical significance is given by a function called probability density or probability function.

Time Independent Schrödinger Wave Equation:

Consider a particle of mass 'm' moving with velocity 'v'. The de-Broglie wavelength 'λ' is

$$\lambda = \frac{h}{mv} = \frac{h}{p} \longrightarrow (1)$$

Where $p=mv$ is the momentum of the particle.

The de- Broglie wave equation for a particle travelling in positive x-axis can be written in complex form as

$$\Psi = A\, e^{i(kx-\omega t)} \longrightarrow (2)$$

Where 'A' is amplitude of the wave and 'ω' is the angular frequency of the wave, Ψ is the Total wave function and k is the wave number.

The space independent term in an equation (2) can be represented as a wave function,

$$\Psi = A\, e^{i(kx)} e^{i(-\omega t)}$$

$$\Psi = \psi e^{-i\omega t} \longrightarrow (3)$$

where $\psi = A\, e^{i(kx)}$

Differentiating equation (3) with respect to 'x' twice, we get

$$\frac{d^2\Psi}{dx^2} = e^{-i\omega t}\frac{d^2\psi}{dx^2} \longrightarrow (4)$$

Differentiating equation (3) with respect to 't' twice, we get

$$\frac{d\Psi}{dt} = (-i\omega)\, e^{-i\omega t}\psi$$

$$\frac{d^2\Psi}{dt^2} = (-i\omega)(-i\omega)e^{-i\omega t}\psi$$

$$\frac{d^2\Psi}{dt^2} = (i^2\omega^2)e^{-i\omega t}\psi$$

$$\frac{d^2\Psi}{dt^2} = -\omega^2\, e^{-i\omega t}\psi \qquad \longrightarrow (5)$$

We know that, the equation for a travelling wave in differential form can be written as

$$\frac{d^2\mathrm{y}}{dx^2} = \frac{1}{v^2}\frac{d^2y}{dt^2}$$

Then by analogy, the wave equation for a de Broglie wave for the motion of free particle propagating in x-axis with a velocity *v*, with a displacement *x* at a time t, can be written as

$$\frac{d^2\Psi}{dx^2} = \frac{1}{v^2}\frac{d^2\Psi}{dt^2} \quad \longrightarrow (6)$$

Substituting eqn. (4) and (5) in the above equation (6), we get

$$e^{-i\omega t}\frac{d^2\psi}{dx^2} = \left(\frac{1}{v^2}\right) - \omega^2\, e^{-i\omega t}\psi$$

$$\frac{d^2\psi}{dx^2} = \frac{-\omega^2}{v^2}\,\psi$$

But we know that, $\omega = 2\pi\nu$ and $v = \nu\lambda$. Substitute these values in the above equation.

$$\frac{d^2\psi}{dx^2} = \frac{-4\pi^2\nu^2}{\nu^2\lambda^2}\psi = \frac{-4\pi^2}{\lambda^2}\psi \qquad \longrightarrow (7)$$

From eqn. (1) substitute for λ in the above equation.

$$\frac{d^2\psi}{dx^2} = \frac{-4\pi^2m^2v^2}{h^2}\psi$$

$$\frac{d^2\psi}{dx^2} + \frac{4\pi^2m^2v^2}{h^2}\psi = 0 \qquad \longrightarrow (8)$$

We know that, Total Energy = Kinetic Energy + Potential Energy

$$E = \frac{1}{2}\,mv^2 + V$$

Multiply both sides by *2m*, we get

$$2mE = \ m^2v^2 + 2mV$$

$$m^2v^2 = 2m\ (E - V) \quad \longrightarrow (9)$$

Substitute eqn. (9) in the equation (8), then we get

$$\frac{d^2\psi}{dx^2} + \frac{4\pi^2(2m).(E-V)}{h^2}\psi = 0$$

$$\boxed{\frac{\boldsymbol{d^2\psi}}{\boldsymbol{dx^2}} + \frac{\boldsymbol{8\pi^2m\ (E-V)}}{\boldsymbol{h^2}}\boldsymbol{\psi} = \boldsymbol{0}}$$

This is known as time independent Schrodinger wave equation in 1-Dimension.

$$\boxed{\frac{\boldsymbol{d^2\psi}}{\boldsymbol{dx^2}} + \frac{\boldsymbol{d^2\psi}}{\boldsymbol{dy^2}} + \frac{\boldsymbol{d^2\psi}}{\boldsymbol{dz^2}} + \frac{\boldsymbol{8\pi^2m\ (E-V)}}{\boldsymbol{h^2}}\boldsymbol{\psi} = \boldsymbol{0}}$$

This is known as time independent Schrodinger wave equation in 1-Dimension.